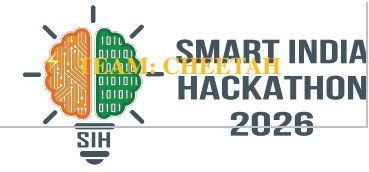
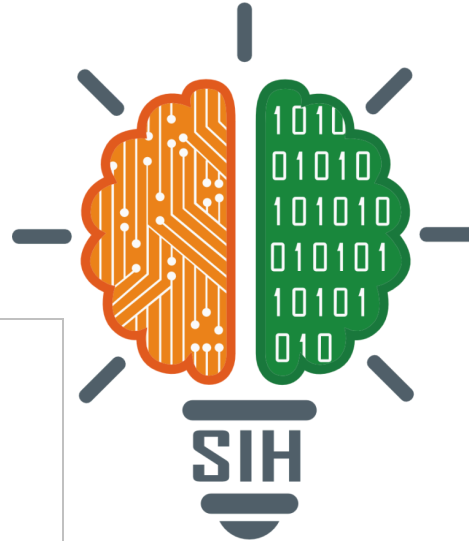
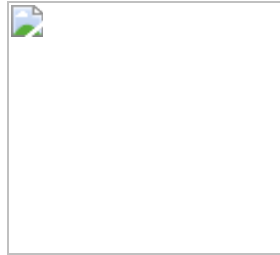


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** – SIH 2026 (PS-1736)
- **Problem Statement Title** – MetraVerify: Digital Verification & Certification System for Weighing and Measuring Instruments (Legal Metrology)
- **Theme** – Smart Automation / Public Governance & Legal Metrology
- **PS Category** – Software
- **Team ID** – SIH2026-CHEETAH
- **Team Name** – CHEETAH



EVERY TRANSACTION RESTS ON A SCALE

Verification exists in law, but trust is still kept in files, registers and memory



The gap today

The Problem: Legal Metrology

- **Applications and records** on paper or in isolated local systems
- **No single view** of what is verified, pending or overdue
- **Expiry** tracked by memory, so lapses go unnoticed
- **Consumers** cannot check a stamp themselves

Who it hurts

- **Customers:** short weight, wrong quantity
- **Honest traders:** penalised for missed renewals
- **Officers:** paperwork instead of inspection
- **Government:** no real-time enforcement picture

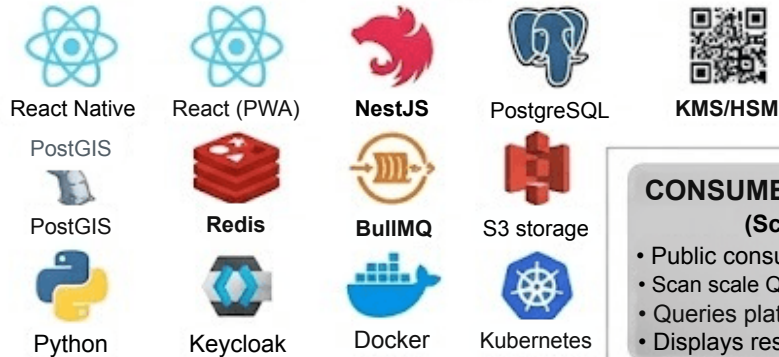
Our vision

- One national digital trust layer for Legal Metrology enforcement.
- Central Authority Triage: Admin dispatches to local inspectors.
 - Automated local
- Fair markets, honest business, empowered 1.4B consumers.
- From inspector matching by applicant to applicant matching by inspector
- From manual paper stamps to real-time tamper-evident QR verification.
- From manual district & pincode.

SYSTEM PRINCIPLES & COMPONENTS

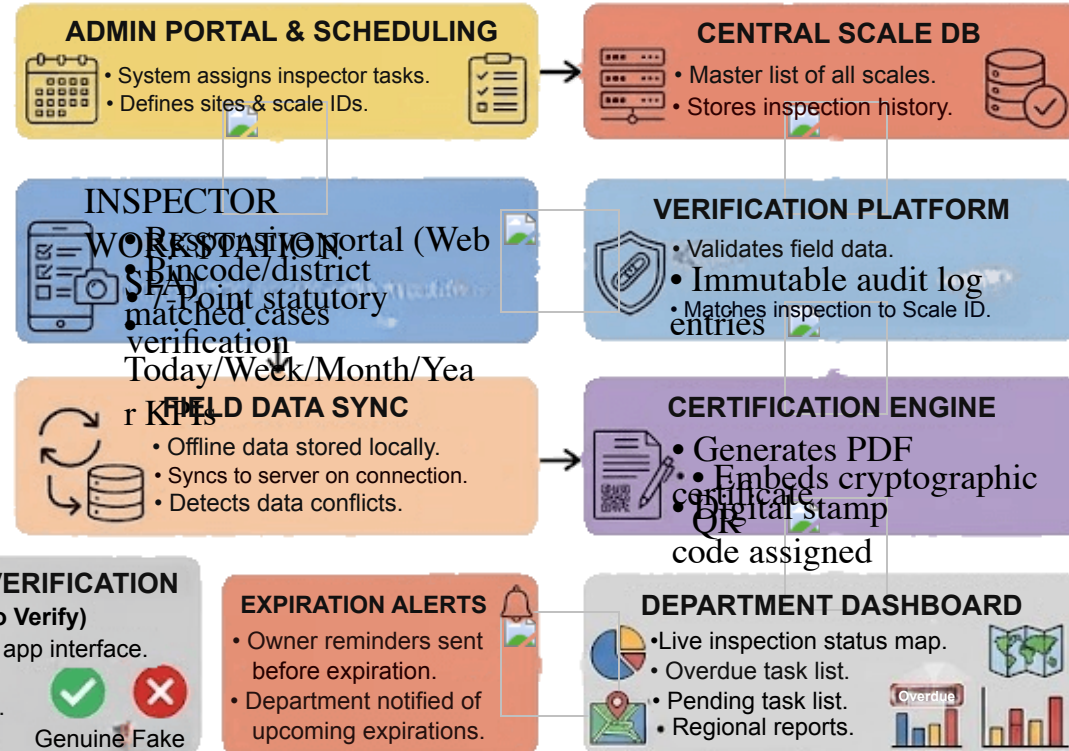
- Central Authority triage & inspection allocation.
- 7-Point Statutory Digital Inspection Engine (visual, zero, seamless data synchronization to central server.
- Role-Based Access Control (RBAC) across 4 Portals: Public Digital Audit LMD, & ATC stamps for Tamper Evident QR Verification with clean UI/UX for public verification (no package).
- Live National Stamping & Verification Throughput analytics.

Test Stack: AKRS Prototype (Live) + Scaled Architecture

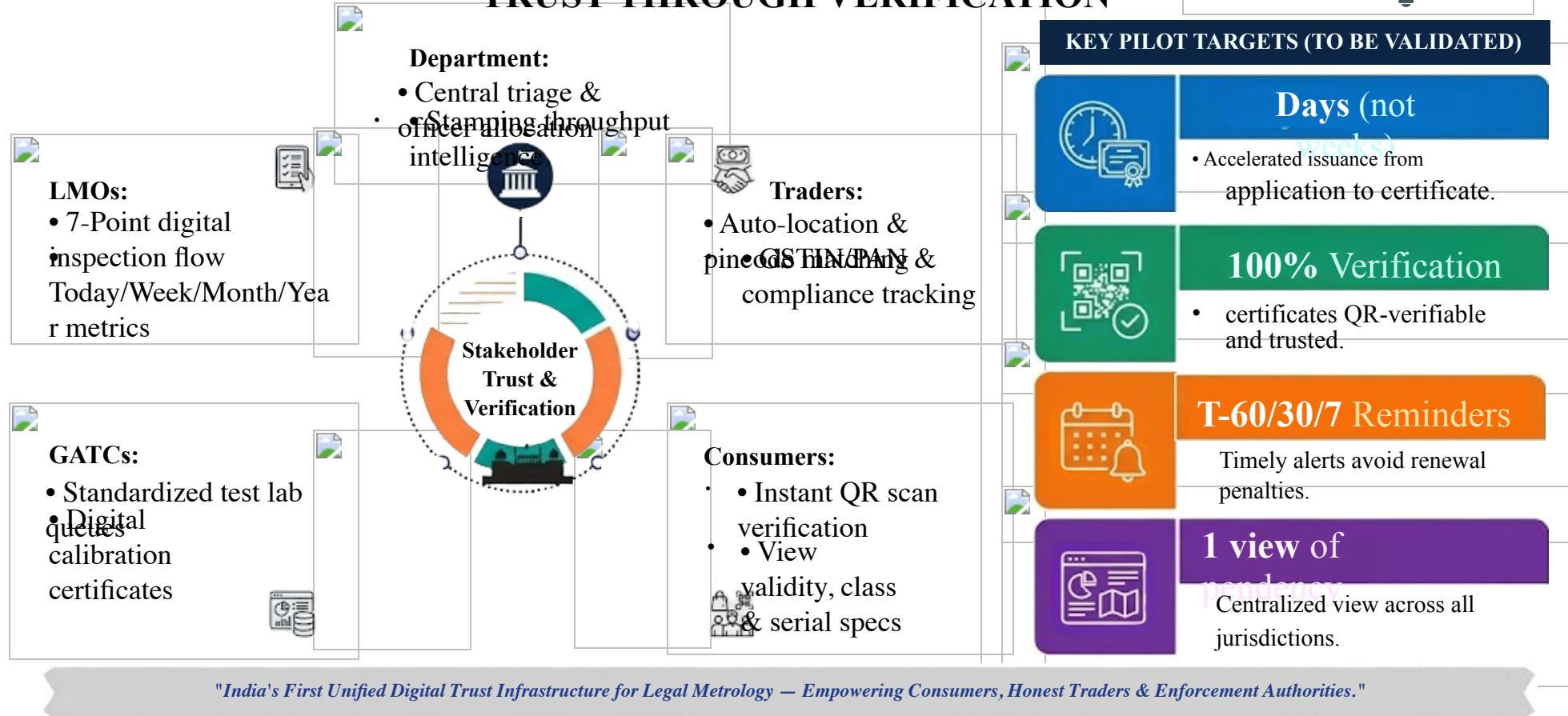


SCALE VERIFICATION WORKFLOW

SCALE CHECK SYSTEM

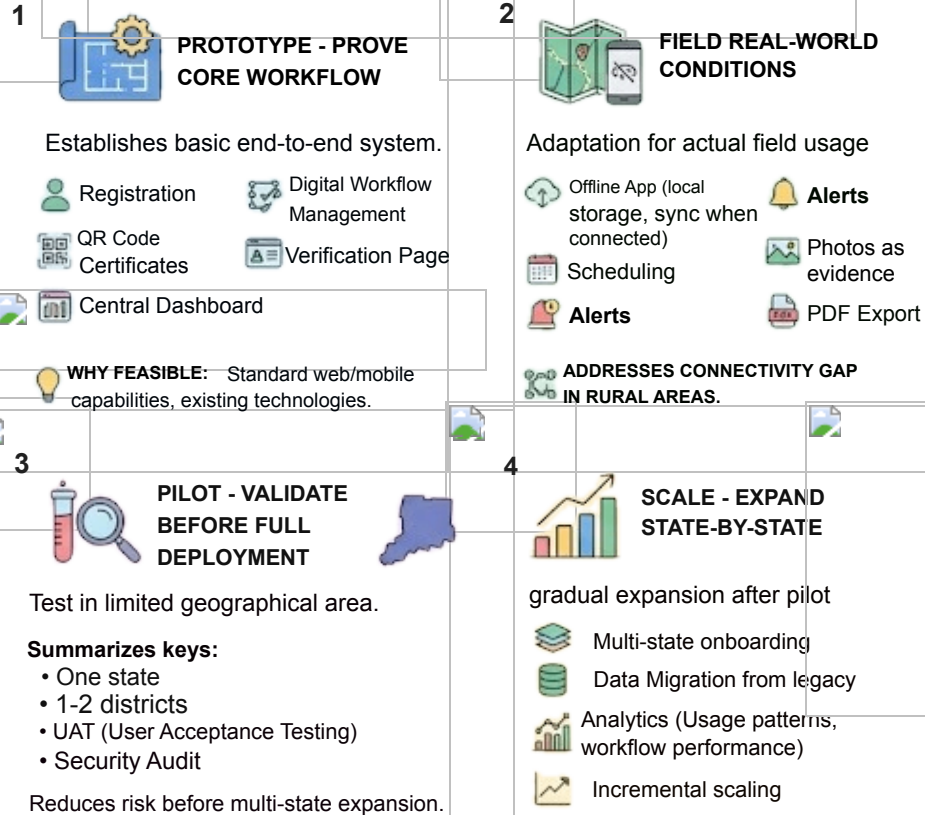


STAKEHOLDER IMPACT & BENEFITS: TRUST THROUGH VERIFICATION



PROJECT FEASIBILITY & ROLLOUT STRATEGY: PROTOTYPE TO NATIONWIDE SCALE

ROLLOUT PHASES: INCREMENTAL GROWTH STRATEGY



KEY RISKS & MITIGATION STRATEGIES:

PROACTIVE FEASIBILITY ADDRESS

 RISK	 SOLUTION/MITIGATION
POOR FIELD CONNECTIVITY: Use of limited internet in remote areas.	OFFLINE-FIRST APP, AUTO-SYNC, CONFLICT RULES. Local storage, predefined resolution rules.
LOW DIGITAL LITERACY: Users not comfortable with complex systems.	MOBILE-FIRST, REGIONAL LANGUAGES, OTP SMS/WHATSAPP, HELPDISK. Adoptions as a *design* problem.
CERTIFICATE FORGERY: Certificates altering, or falsely re-issuing.	SIGNED QR, PUBLIC VERIFICATION, SERIAL-LINKED PHYSICAL STAMP. Digital +Physical traceability combined.
STATE-WISE RULE DIFFERENCES: Varying laws, fees, jurisdictions, structures differ.	CONFIGURABLE WORKFLOW, FEE MASTERS, JURISDICTION MASTERS. Common platform, configurable rules.
PRIVACY & SECURITY: Handling sensitive user/admin information.	ROLE-BASED ACCESS CONTROL (RBAC), AUDIT TRAILS, DPDP-ALIGNED DESIGN, VAPT. Accountability and risk check.



RESEARCH AND REFERENCES



- **1. Legal Metrology Act, 2009 & General Rules, 2011:** Statutory framework governing verification, inspection checklists, and mandatory stamping periods under the Department of Consumer Affairs, Govt. of India.
- 2. OIML International Recommendation R 76-1:** Metrological and technical specifications for Non-Automatic Weighing Instruments (Accuracy Classes I, II, III, IIII, eccentricity, and repeatability tests).
- 3. Bureau of Indian Standards (BIS) & NPL Traceability:** Guidelines on reference standards, laboratory calibration hierarchy, and GATC verification protocols.
- 4. Department of Consumer Affairs E-Commerce Guidelines:** Mandates on digital stamping display and consumer verification for commercial trade transactions.
- 5. Live Working MetraVerify Prototype & Codebase:** Production-tested MERN architecture: <https://github.com/Pushkar-8950/Ecommerce-Frontend>

IMPORTANT INSTRUCTIONS



Please ensure below pointers are met while submitting the Idea PPT:

1. Kindly keep the maximum slides limit up to six **(6)**. (Including the title slide)
2. Try to avoid paragraphs and post your idea in points /diagrams / Infographics /pictures
3. Keep your explanation precise and easy to understand
4. Idea should be unique and novel.
5. You can only use provided template for making the PPT without changing the idea details pointers (mentioned in previous slides).
6. You need to save the file in PDF and upload the same on portal. No PPT, Word Doc or any other format will be supported.

Note - You can delete this slide (Important Pointers) when you upload the details of your idea on SIH portal.